Helper Functions API

ASTR 3070 Fall 2026

fetch_filenames

`fetch_filenames(dir)`

Fetches a list of filenames from a directory.

Args:
dir (str): path of the file directory

Returns:
list[str]: list of filenames

stack_master

`stack_master(files)`

Stacks frames to create a master image.
This does NOT align the frames when stacking.

Args:
files (list): list of filenames

Returns:
(np.array): final image stack in the form of a 2D array

calibrate_lights

`calibrate_lights(lights, dark, flat_bias)`

Applies dark and bias/flat corrections to individual light frames.

Args:
lights (list[str]): list of light frames
dark (np.array): master dark frame
flat_bias (np.array): master flat/bias frame

Returns:
list[np.array]: list of corrected light frames

align_images

`align_images(nights, ref_image)`

Aligns images from multiple nights.

Args:
nights (list[list[np.array]]): list of calibrated sequences from each night
ref_image (np.array): image to use as reference for alignment

Returns:
(np.array): aligned master stack array
count (int): returns the number of frames stacked